

V2: Directional Forcing

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When does a second wave direction break the single-source inverse?

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SHARC buoy project Code: `runDirectionalForcingStudy.m`, `v2Tools.m`, `directionalSpectralMatrix.m`

`diagnoseV2CoherenceNoise.m` Results: `Results/V2/V2.20261005_142206` (commit 2e163b1)

Summary

Question. The output-only inverse rests on one incident wave component: then the spectral matrix is rank one and $T_j = S_{jr}/S_{rr} = H_j/H_r$. A second, independent direction makes it rank two and the measured ratio becomes a mixture the frozen inverse reads as a change of β and R . How large is that bias, and does the coherence loss, which is observable, warn before it matters? **Outcome.** The single-direction inverse is extremely sensitive to secondary energy. All 16 registered cases ($\phi = 15$ to 90° , $f_2 = 5$ to 50%) are Not identifiable, with $d_{\text{sys}} = 7.5$ to 1342 . The useful numbers are the small-contamination crossings: $d_{\text{sys}} = 0.5$ is reached at **0.33%** of the incident energy at 15° separation, falling to **0.022%** at 90° , in a regime verified to be linear (and nonlinearly at $d_{\text{sys}} = 1$). Coherence warns before the bias matters only for well-separated directions (60 and 90°), and only if the in-band noise PSD is known to a few percent: underestimating it by 4.4% already produces false alarms. Directional forcing is a structural model mismatch, not an estimator failure.

1 What the study does

Why V2 exists. Every earlier study (and the G3 inverse) assumes one coherent forcing direction. V2 violates that assumption on purpose while the inverse stays frozen, to find how much directional contamination it tolerates.

Why one direction matters. With one component, $\mathbf{y} = \mathbf{H}_1 u_1$ and $S_{jr} = H_{1j} H_{1r}^* S_1$, $S_{rr} = |H_{1r}|^2 S_1$, so $S_{jr}/S_{rr} = H_{1j}/H_{1r}$ exactly: the input spectrum cancels. With two independent components, $\mathbf{y} = \mathbf{H}_1 u_1 + \mathbf{H}_2 u_2$,

$$\frac{S_{jr}}{S_{rr}} = \frac{w_1 T_{1j} + w_2 T_{2j}}{w_1 + w_2}, \quad w_c = |H_{cr}|^2 S_c,$$

a mixture of the two directions' transmissibilities weighted by the *reference sensor's response energy* to each, not by the incident energy fractions alone. The signal spectral matrix is rank two unless the two response vectors are collinear, and no choice of (β, R) in the single-direction model reproduces it. The inverse can therefore converge perfectly and still be biased.

2 Method

Pre-registered in `cfg.v2`; a clarification and one amendment were logged before any V2 run.

1. **Forcing.** Primary component at 0° (the EMM incidence, the Level 2C axis), secondary at $+\phi$, both with the reference JONSWAP shape; total energy fixed, $S_{u1} = (1 - f_2)S_u$, $S_{u2} = f_2 S_u$, so directionality is not confounded with SNR. For a circular floe a component from heading ϕ is the nominal field at the sensors rotated by $-\phi$; one EMM field at the rotated Level 2C points serves every case. Sweep $\phi \in \{15, 30, 60, 90\}^\circ$, $f_2 \in \{0.05, 0.10, 0.25, 0.50\}$ at p_0 , Level 2C, reference s26, LSM6DSV16X, setting F.
2. **Spectral matrix.** $\mathbf{S} = \sum_c \mathbf{H}_c \mathbf{H}_c^H S_{u,c} + \text{diag}(S_{nn,j})$, smoothed with the engine's Welch kernel (one shared function, `directionalSpectralMatrix`, also used by D3). Both the expected transmissibility and its covariance come from this rank-two matrix through the unchanged engine functions.
3. **Bias.** $b_{\text{dir}} = f_W^{\text{dir}} - f_W^{\text{single}}$, both Welch expectations through the same path, projected with the nominal J, Σ, F : $\delta\theta_{\text{dir}} = F^{-1} J^T \Sigma^{-1} b_{\text{dir}}$, $d_{\text{sys}} = \sqrt{\delta\theta_{\text{dir}}^T F \delta\theta_{\text{dir}}}$; the class uses the vector sum $\delta\theta_W + \delta\theta_{\text{dir}}$ under the nominal σ .
4. **Crossings.** Per ϕ , the first f_2 at which $d_{\text{sys}} = 0.5, 1$ and $|\delta\beta/\beta| = 1\%$ are reached, by bisection in $\log_{10} f_2$ inside the bracket between the exact $f_2 = 0$ baseline and the first registered point.
5. **Coherence.** $\gamma_{jr}^2 = |S_{jr}|^2 / (S_{jj} S_{rr})$ per pair, Fisher $z = \text{atanh}|\gamma|$. The expected drop of the band-mean z from the single-direction baseline is divided by its null scatter, *measured* by Monte Carlo of the real estimator (400 single-direction records; amendment: the bins are not assumed independent). Detectable when $\max_j D_j \geq z_{1-0.05/3} + z_{0.80} = 2.97$, the 80%-power threshold (the 5% rejection threshold is 2.13). Coherence warns early at ϕ if $f_{\text{detect}} < f_2(d_{\text{sys}} = 0.5)$.
6. **Checks.** Tester 13/13: the single-direction path reproduces the engine (f_W to 10^{-14}), $\pm\phi$ mirror symmetry on the EMM, rank and coherence, energy conservation, local linearity. Nonlinear G3 inversion at the most sensitive $d_{\text{sys}} = 1$ crossing.

3 Results

3.1 The registered 5 to 50% grid

ϕ	d_{sys}				$\delta\beta/\beta$			
	$f_2 = 0.05$	0.10	0.25	0.50	0.05	0.10	0.25	0.50
15°	7.55	15.1	38.1	77.0	+2.7%	+5.4%	+14%	+31%
30°	29.2	58.8	150	313	+10%	+22%	+67%	+190%
60°	85.0	173	452	988	+23%	+53%	+206%	+1063%
90°	117	238	620	1342	-2.9%	-6.1%	-17%	-39%

All 16 Not identifiable. Linear projections: beyond $d_{\text{sys}} \sim 1$ the parameter biases are qualitative (V1). The scatter grows too: $\sigma_{\ln \beta}$ under the directional sea is 1.34, 2.77, 4.26 and 2.85 times nominal at $f_2 = 0.5$.

Even 5% secondary energy breaks the inverse, and the damage grows with separation. At 90° the bias goes mostly into R ($\delta R/R = -4.6\%$ against $\delta\beta/\beta = -2.9\%$ at 5%), so the Fisher distance is largest there while the stiffness error is not.

3.2 Small contamination: the tolerances

The bias is linear in f_2 : d_{sys}/f_2 is constant to 0.02% over $f_2 = 2.5 \times 10^{-4}$ to 10^{-3} (slopes 151, 579, 1677 and 2313 for 15, 30, 60, 90°) and still within 1% at $f_2 = 0.05$. The crossings therefore lie in a genuinely local regime.

ϕ	f_2 at $d_{\text{sys}} = 0.5$	$d_{\text{sys}} = 1$	$ \delta\beta/\beta = 1\%$	f_{detect} (coherence)	early warning
15°	0.332%	0.664%	1.88%	2.01%	no
30°	0.086%	0.172%	0.51%	0.134%	no
60°	0.030%	0.060%	0.24%	0.0129%	yes ($\times 2.3$)
90°	0.022%	0.043%	1.75%	0.0063%	yes ($\times 3.4$)

Fractions of the total incident energy. Early warning: $f_{\text{detect}} < f_2(d_{\text{sys}} = 0.5)$, factor = $f_2(d_{\text{sys}} = 0.5)/f_{\text{detect}}$.

Statistical inconsistency arrives long before a large physical error: at 90°, $d_{\text{sys}} = 1$ is reached when the stiffness bias is only 0.075%, and 1% in β needs about 80 times the energy that gives $d_{\text{sys}} = 0.5$.

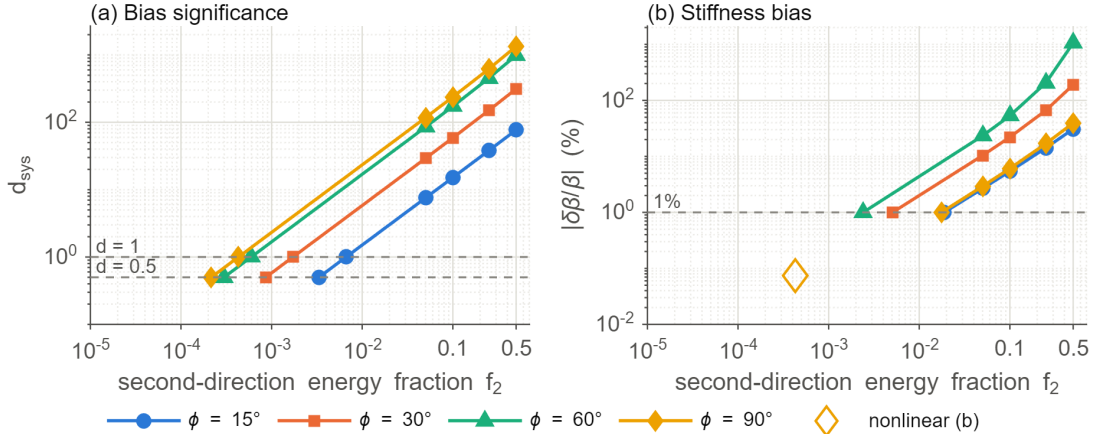


Figure 1. Inverse bias against the second-direction energy fraction, per separation ϕ : (a) d_{sys} with the 0.5 and 1 limits; (b) $|\delta\beta/\beta|$ with the 1% limit. Registered points and the bisected crossings; open marker: the nonlinear inversion at the 90°, $d_{\text{sys}} = 1$ crossing.

3.3 Nonlinear check

At $\phi = 90^\circ$, $f_2 = 4.31 \times 10^{-4}$ ($d_{\text{sys}} = 0.997$) the G3 solver converged in 2 iterations (8 model evaluations) to $\delta\beta/\beta = +0.075\%$, $\delta R/R = -0.0295\%$, equal to the linear projection (gap 0.000σ). The tolerances are therefore not a linearisation artefact, and the failure is not an optimiser pathology: the solver finds the wrong answer cleanly.

3.4 Coherence as a validity diagnostic

Single-direction pair coherence is about 0.99 over the band. The measured null scatter of the band-mean Fisher z was $[0.0103, 0.0111, 0.0103]$ against 0.0101 for independent bins (variance ratios 1.05, 1.22, 1.04; effective bins 69 to 81 of 84), so the independence assumption would have been mildly optimistic; the calibrated values are used. Coherence warns first at 60 and 90° but not at 15 and 30° (table above): close headings give nearly collinear response vectors, the spectral matrix stays nearly rank one, yet the very precise inverse is already biased.

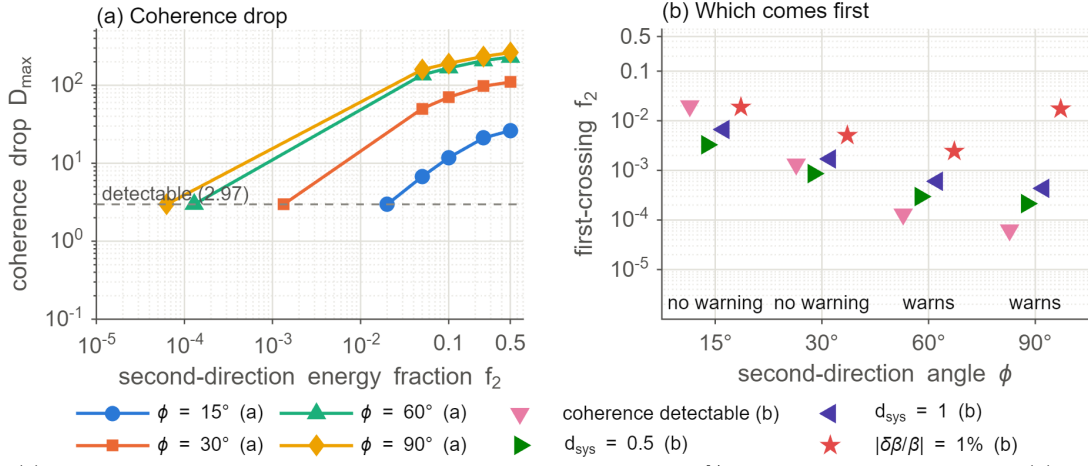


Figure 2. (a) Expected standardised coherence drop $D_{\max}$ against f_2 , with the 80%-power detectability threshold. (b) First-crossing fractions per separation: coherence warns early where its marker lies below the $d_{\text{sys}} = 0.5$ marker.

Noise-floor sensitivity (post-run diagnostic, not registered). The baseline coherence is set mostly by the noise floor, so an error ϵ in the assumed S_{nn} shifts it by about $\epsilon/2$ in Fisher z per bin.

ϕ	f_{detect} with the baseline at $S_{nn}(1 + \epsilon)$					$f_2(d_{\text{sys}} = 0.5)$
	$\epsilon = -10\%$	-5%	0	$+5\%$	$+10\%$	
15°	false alarm	0.433% (late)	2.01% (late)	3.68% (late)	5.44% (late)	0.332%
30°	false alarm	0.029% (early)	0.134% (late)	0.243% (late)	0.353% (late)	0.086%
60°	false alarm	0.0028% (early)	0.0130% (early)	0.0233% (early)	0.0342% (late)	0.030%
90°	false alarm	0.0014% (early)	0.0063% (early)	0.0115% (early)	0.0169% (early)	0.022%

With no second direction, an underestimated noise floor alone reaches the 5% rejection threshold at $\epsilon = -4.39\%$ and the 80%-power threshold at -6.07% ; an overestimate masks real drops.

The diagnostic is about five times stricter on the noise floor than the inverse itself (R3 required $\pm 30.6\%$), and the risk is asymmetric: underestimating S_{nn} manufactures directional detections.

4 Interpretation

- The failure is structural.** Under independent directions $S_{jr}/S_{rr} \neq H_j/H_r$; the frozen inverse converges and returns biased (β, R). No estimator tuning removes this.
- Tiny secondary energy is statistically decisive** because the nominal inverse is so precise ($\sigma_{\ln \beta} = 0.36\%$): 0.02 to 0.3% of the incident energy reaches $d_{\text{sys}} = 0.5$.
- Fisher significance and physical error answer different questions.** At 90° the mismatch is mostly in R ; d_{sys} flags it 80 times earlier than the 1% stiffness criterion.
- Coherence is useful but conditional.** It warns early for well-separated directions only, and only with the in-band noise PSD calibrated to a few percent; it cannot certify the single-direction assumption for close headings.
- This motivates D3:** V2 identifies the failure at the Level 2C layout; D3 asks whether sensor count and placement can make the inverse robust to it.

Conclusion. V2 deliberately violated the single-input assumption by adding a second independent incident direction while the inverse remained frozen to a single 0° input. The resulting transmissibility mismatch was extremely influential: the $d_{\text{sys}} = 0.5$ tolerance fell from about 0.33% secondary energy at 15° separation to 0.022% at 90° . These small-fraction crossings lie in a strictly linear regime, and a full nonlinear inversion at the most sensitive $d_{\text{sys}} = 1$ crossing agreed exactly with the local prediction. Coherence gave an early warning at 60° and 90° but not at 15° or 30° , and only when the noise floor is known to within a few percent. Coherence is therefore a useful diagnostic for strongly separated independent forcing, but cannot by itself certify the single-direction model.

Limitations. One floe (p_0), the Level 2C layout and one sea shape; two ideal, fully independent components (a real sea is spread and partly coherent; D3b adds a spread case); four separations only. f_2 is an incident-energy fraction, while the effective mixing is weighted by the reference response. The class uses the nominal σ (the frozen inverse's ruler); the actual scatter under the directional sea is larger. The coherence baseline is the model prediction at the true parameters; the noise diagnostic is post-run. Apart from the forcing, truth and inverse share the EMM.

Requirement for the experiment and the thesis

The single-direction transmissibility inverse must not be applied blindly under multidirectional forcing: secondary energy of order 0.1% of the total can bias it beyond its own uncertainty. For well-separated secondary directions, coherence can serve as an early warning only when the in-band reference noise PSD is independently calibrated to within a few percent (underestimation creates false alarms); closely separated components can bias the inverse before any coherence warning. Directional forcing must therefore be modelled explicitly or treated as a validity condition of the frozen inverse. The Conclusion above is the thesis wording.

Outputs (Results/V2/): V2.20261005_142206.log/.mat/_summary.txt; V2_grid.csv, V2_crossings.csv, V2_nonlinear.csv; diagnostics/V2noise_20261005_150224.mat, V2_noise_falseAlarm.csv, V2_noise_fdetect.csv; figures/V2_fig1, V2_fig2 (.pdf/.png) from plotDirectionalForcingStudy. Tester testDirectionalForcingStudy 13/13; run 374 s.